

Groups and Lie Algebras: Exercise Sheet 8

1 Gell–Mann matrices of $\mathfrak{sl}(3)$ The *Gell–Mann matrices* are the eight Hermitian, traceless 3×3 matrices

$$\begin{aligned}\lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda_4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda_5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.\end{aligned}$$

They satisfy $\text{tr } \lambda_a = 0$ and $\text{tr}(\lambda_a \lambda_b) = 2\delta_{ab}$, so the matrices $\frac{1}{2}\lambda_a$ form an orthonormal basis (w.r.t. $\langle X, Y \rangle = \frac{1}{2} \text{tr}(XY)$) of the real vector space of Hermitian traceless matrices, i.e. of $\mathfrak{su}(3)$ after multiplication by i .

Let $E_{ij} \in \mathfrak{gl}(3, \mathbb{C})$ be the *matrix unit* with entries

$$(E_{ij})_{kl} = \delta_{ik} \delta_{jl},$$

i.e. a single 1 in row i , column j , and zeros elsewhere.

1. Show that $E_{ij}E_{kl} = \delta_{jk} E_{il}$, and deduce the general commutator

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{li} E_{kj}.$$

2. The traceless matrices form $\mathfrak{sl}(3, \mathbb{C})$. Check that the six off-diagonal units E_{ij} ($i \neq j$), together with

$$H_{12} = E_{11} - E_{22}, \quad H_{23} = E_{22} - E_{33},$$

form a basis. Express the Gell–Mann ladder combinations $\frac{1}{2}(\lambda_1 \pm i\lambda_2)$, $\frac{1}{2}(\lambda_4 \pm i\lambda_5)$, $\frac{1}{2}(\lambda_6 \pm i\lambda_7)$ in terms of the E_{ij} .

2. Tensor products of $\mathfrak{sl}(3)$ representations Decompose the tensor product of the 8-dimensional adjoint representation of $\mathfrak{sl}(3)$ with the defining three dimensional representation on $\mathbb{C}^3$ into irreducible representations.

3. The Weyl group of $\mathfrak{sl}(3)$ and invariance of weight diagrams Recall the Cartan subalgebra $\mathfrak{h} = \langle H_{1,2}, H_{2,3} \rangle$ of $\mathfrak{sl}(3, \mathbb{C})$ and consider the two roots $\alpha_1, \alpha_2 \in \mathfrak{h}^*$ given by $\alpha_1 = L_1 - L_2$, $\alpha_2 = L_2 - L_3$. The *simple reflections* s_1, s_2 are the linear maps on the span of α_1, α_2 defined by

$$s_1 : \quad \alpha_1 \mapsto -\alpha_1, \quad \alpha_2 \mapsto \alpha_1 + \alpha_2, \quad s_2 : \quad \alpha_1 \mapsto \alpha_1 + \alpha_2, \quad \alpha_2 \mapsto -\alpha_2,$$

Equivalently, on a general element $\mu = a\alpha_1 + b\alpha_2$,

$$s_1(\mu) = (b - a)\alpha_1 + b\alpha_2, \quad s_2(\mu) = a\alpha_1 + (a - b)\alpha_2.$$

The *Weyl group* is the group generated by those two reflections $W = \langle s_1, s_2 \rangle$.

1. Draw the action of the Weyl group on the root lattice, verify $s_1^2 = s_2^2 = e$ and $(s_1 s_2)^3 = e$, and conclude that W is a group of order 6 isomorphic to S_3 .
2. Let V be any finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$, with weight space decomposition $V = \bigoplus_{\mu} V_{\mu}$, where $V_{\mu} = \{v \in V : H \cdot v = \mu(H)v \text{ for all } H \in \mathfrak{h}\}$. Fix the roots α_1 and α_2 consider the corresponding $\mathfrak{sl}(2)$ sub-Lie algebras $\{E_{1,2}, E_{2,1}, H_{1,2}\}$ and $\{E_{2,3}, E_{3,2}, H_{2,3}\}$ of $\mathfrak{sl}(3)$. Using the representation theory of $\mathfrak{sl}(2, \mathbb{C})$, show that

$$\dim V_{\mu} = \dim V_{s_{\alpha}(\mu)}.$$

Conclude that the set of weights of V , counted with multiplicity, is invariant under every s_i , hence under all of W .

3. As a check, apply part (3) to the adjoint representation $V = \mathfrak{sl}(3, \mathbb{C})$ and to the fundamental representation $V = \mathbb{C}^3$, and describe the W -orbits of weights in each case.
4. Is there a larger symmetry group which acts on all weight diagrams?

4. Sorting hadrons into SU(3) multiplets The table below lists seventeen hadrons in no particular order, with electric charge Q , strangeness S , baryon number B , and mass (in MeV/c^2 , approximate).

Particle	Q	S	B	Mass (in MeV/c^2)
Σ^0	0	-1	1	1193
K^+	+1	+1	0	494
n	0	0	1	940
π^-	-1	0	0	140
Ξ^-	-1	-2	1	1322
η	0	0	0	548
Σ^+	+1	-1	1	1189
K^0	0	+1	0	498
Λ	0	-1	1	1116
π^+	+1	0	0	140
Ξ^0	0	-2	1	1315
η'	0	0	0	958
p	+1	0	1	938
K^-	-1	-1	0	494
Σ^-	-1	-1	1	1197
π^0	0	0	0	135
$\bar{K}^0$	0	-1	0	498

1. Try first to organise the particles by mass alone. Explain why this fails to reveal any clean multiplet structure.
2. Compute $Y = B + S$ (hypercharge which measures the weight of the Gell-Mann matrix $1/\sqrt{3}\lambda_8$) and $I_3 = Q - \frac{1}{2}Y$ (isospin) for each particle. Show that if you plot all seventeen on a single (I_3, Y) diagram they pile up and overlap; conclude that one must first separate the particles by a quantum number that the weight (I_3, Y) does not see. Which column does the job?
3. Split the table accordingly and draw the corresponding multiplets.
4. In one case you should find *three* particles at the origin $(I_3, Y) = (0, 0)$, but the adjoint diagram **8** accommodates only two (its zero weight has multiplicity 2). Use the decomposition $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$ to argue that one of the three must be the $\text{SU}(3)$ singlet **1**, and explain why the singlet is forced to sit at the origin. Which particle is most naturally assigned to the **1**?